

CMS Data Science Project

1(a) Exploration of the DataSet

This report details an analysis conducted on a dataset from the CMS experiment at the LHC. The dataset contains 225,868 records of jet events, each described by 29 features related to their kinematic and structural properties. The main goal is to classify these jets into two categories: those originating from Higgs boson decays (signal) and those from standard QCD processes (background).

The dataset used in this analysis includes 225,868 records, each with 29 features. These records represent simulated high-energy particle collision events from the CMS experiment at the LHC. Each record corresponds to a single jet produced in proton-proton collisions.

The features describe various physical and kinematic aspects of the jets, such as:

- nSV: the number of secondary vertices.
- jetNTracks: the number of associated particle tracks.
- τ trackEtaRel family: measurements of track pseudorapidity relative to subjet axes.
- τ vertex family and τ flightDistance family: properties related to the location and energy of decay vertices.
- trackSip family: significance values of track impact parameters.

Two columns are used for classification:

- isSignal: indicates if a jet comes from a Higgs boson decay into bottom quarks (value = 1).
- isBackground: identifies jets from ordinary QCD processes (value = 1).

Initial checks of the dataset show that:

- Numerical: All features are numerical (float64), with no categorical features.
- Clean: No missing values are present in any of the columns.
- Well-structured: The column "Unnamed: 0" is an index from the CSV export and will be removed during data preparation.

1(b) Missing Values and Fill Values Analysis

Following an initial assessment of the CMS dataset, a detailed inspection was undertaken to identify both explicit missing values and any implicit placeholder values that might represent undefined or incomplete measurements, a common practice in physics datasets.

The application of `df.isnull().sum()` across all columns confirmed the absence of standard missing values (NaN). This indicates that each data point is technically present within the dataset.

Further scrutiny for suspicious placeholder values revealed the following instances:

- -1 Values: The value -1 was observed 102 times in each of the following features:
 - `tau_vertexEnergyRatio_0`
 - `tau_vertexEnergyRatio_1` These occurrences likely signify instances where the energy ratio associated with a vertex could not be defined or measured, potentially due to issues in vertex reconstruction.
- 0 Values: The value 0 was found 33 times in the `nSV` (number of secondary vertices) feature. While a value of zero could legitimately indicate jets with no detected secondary vertices, its interpretation requires careful consideration within the domain context. It might represent a physically meaningful observation rather than a missing measurement.
- 999 Value: A single instance of the value 999 was identified in the `Unnamed: 0` column. As this column serves as an index and is slated for removal during preprocessing, this isolated value is not a concern for the analytical features.

It is important to note that these placeholder values, although not represented as NaN, can still influence the performance of machine learning models if not appropriately addressed. Depending on the physics interpretation of these values, potential strategies for handling them include:

- Masking: Excluding these data points from the model training process.
- Imputation: Replacing these values with statistically estimated values.
- Feature Engineering: Utilizing these values to create new indicator variables that capture the information about the measurement status.

The use of specific numerical values such as -999, -1, or 0 as placeholders for undefined physics measurements is a recognized convention within the field of high-energy physics and machine learning preprocessing (Paperspace, 2021). These placeholders are routinely employed in experimental datasets to denote instances of undefined measurements or absent detector responses.

1(c) Variable Distribution Comparison using Histograms

To investigate the discriminatory power of each feature in distinguishing between signal and background events, a comparative analysis of their distributions was performed through the generation of overlapping histograms. For each of the 29 features, two distributions were visualized:

- Signal Events ($\text{isSignal}=1$): Representing jets originating from the decay of Higgs bosons into bottom quarks ($H \rightarrow b\bar{b}$).
- Background Events ($\text{isSignal}=0$): Representing jets resulting from standard Quantum Chromodynamics (QCD) processes.

Several methodological considerations were implemented in the creation of these visualizations:

- Normalization: To facilitate a direct comparison of the distributions despite the inherent imbalance in class sizes, all histograms were normalized to probability density rather than absolute event counts.
- Binning Strategy: The bin widths for each histogram were determined using the `numpy.histogram_bin_edges()` function with a consistent number of 50 bins. This choice aimed to achieve a balance between preserving the detailed structure of the distributions and maintaining visual clarity.
- Axis Labelling: Each generated plot was meticulously labelled, with the respective feature name clearly indicated on the x-axis and the corresponding normalized density or its logarithm on the y-axis.

For each feature, the distributional comparison was presented using two distinct vertical scales:

- Linear Scale: This representation provides an intuitive visualization of the primary ranges of values and their relative densities for both signal and background events. It is particularly effective for understanding the central tendencies and overall shapes of the distributions.
 - Advantages: Facilitates straightforward interpretation of common value ranges and effectively visualizes the primary data bulk.
- Logarithmic Scale: This representation serves to emphasize the behavior of the distributions in their tails and to highlight the presence of rare occurrences. This is of particular importance in particle physics, where the identification of infrequent signal events, such as $H \rightarrow b\bar{b}$ decays, necessitates the ability to discern subtle patterns within a potentially high background environment.

- Advantages: Aids in the identification of subtle patterns within low-frequency regions of the data and is crucial in physics for examining tail behaviors, which often correspond to rare but significant high-momentum decays.

Both the linear and logarithmic scale visualizations offer valuable insights into the distributional characteristics of the features for signal and background events. The logarithmic scale is particularly advantageous for detecting potential outliers or subtle, yet crucial, signal patterns that might be obscured by a dense background distribution when viewed on a linear scale. Therefore, the combined interpretation of both perspectives is recommended for a comprehensive understanding of the data.

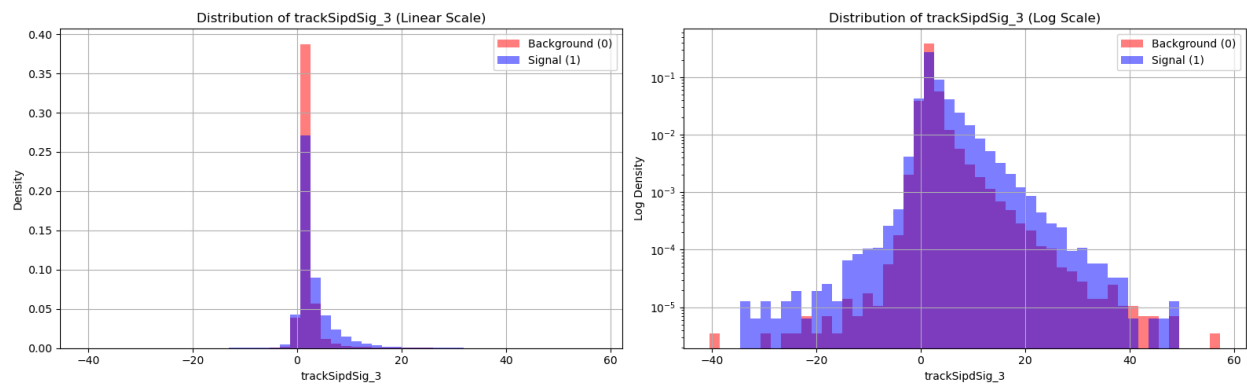


Figure 1: Example histogram plot

1(d) Summary Statistics of Jet Features

To understand the central tendency and variability of each feature, the mean and standard deviation were computed for all numerical columns (excluding label and index columns). These statistics provide insight into the scale and spread of each jet-related measurement.

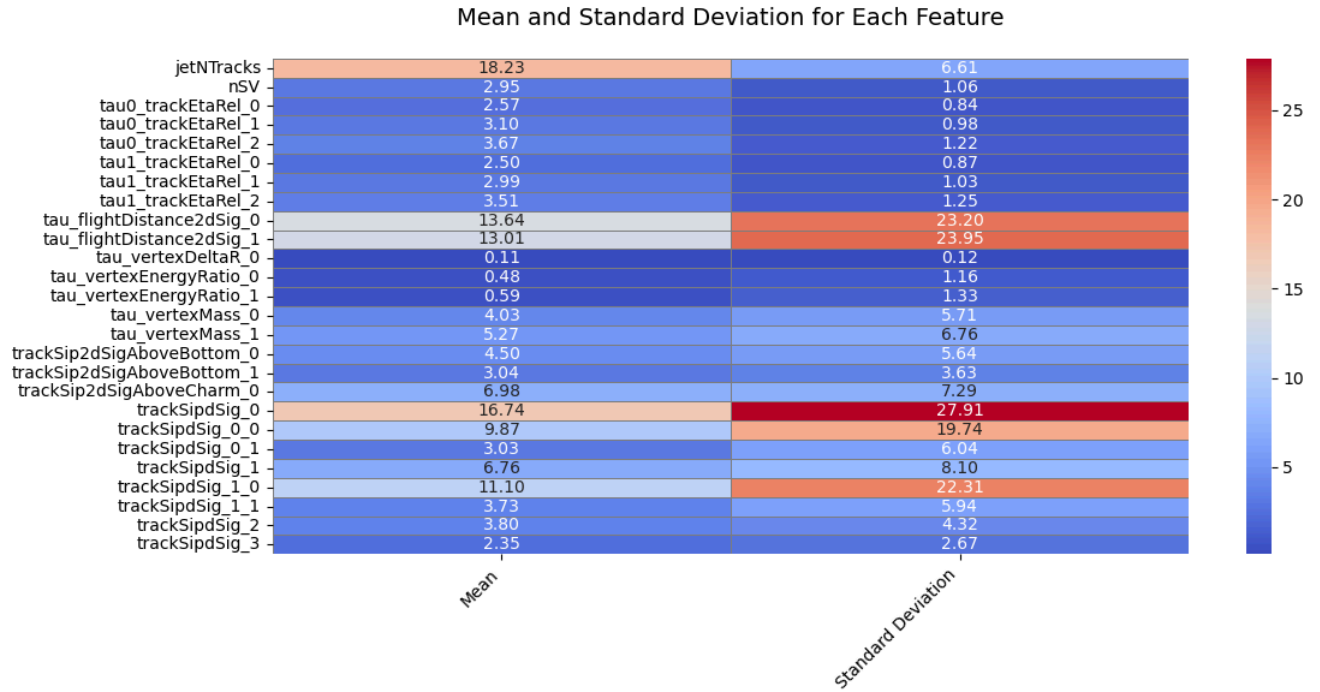


Figure 2: Mean and standard deviation plot

Overall Observations

- Variables related to track impact parameter significance (trackSipdSig_*) tend to have larger means and standard deviations, indicating a wider range of values and greater variability.
- Variables related to energy ratios (tau_vertexEnergyRatio_*) have means around 0.5 but large standard deviations, suggesting a wide spread in their distributions.
- Variables related to vertex distances and masses (tau_flightDistance2dSig_*, tau_vertexMass_*) also exhibit substantial variability.

A heatmap was used to visualize the values across features. This representation allows for quick comparison across different variables and helps detect which features may require scaling or normalization before modeling.

- Features like trackSipdSig_* have large means and standard deviations, suggesting a wide dynamic range.
- Features like tau_vertexEnergyRatio_* show much smaller values, indicating a narrower distribution.

1(e) Correlation Between Numerical Variables

To understand the relationships between different jet features, a correlation matrix was computed using Pearson coefficients. These values range from -1 (strong negative correlation) to +1 (strong positive correlation).

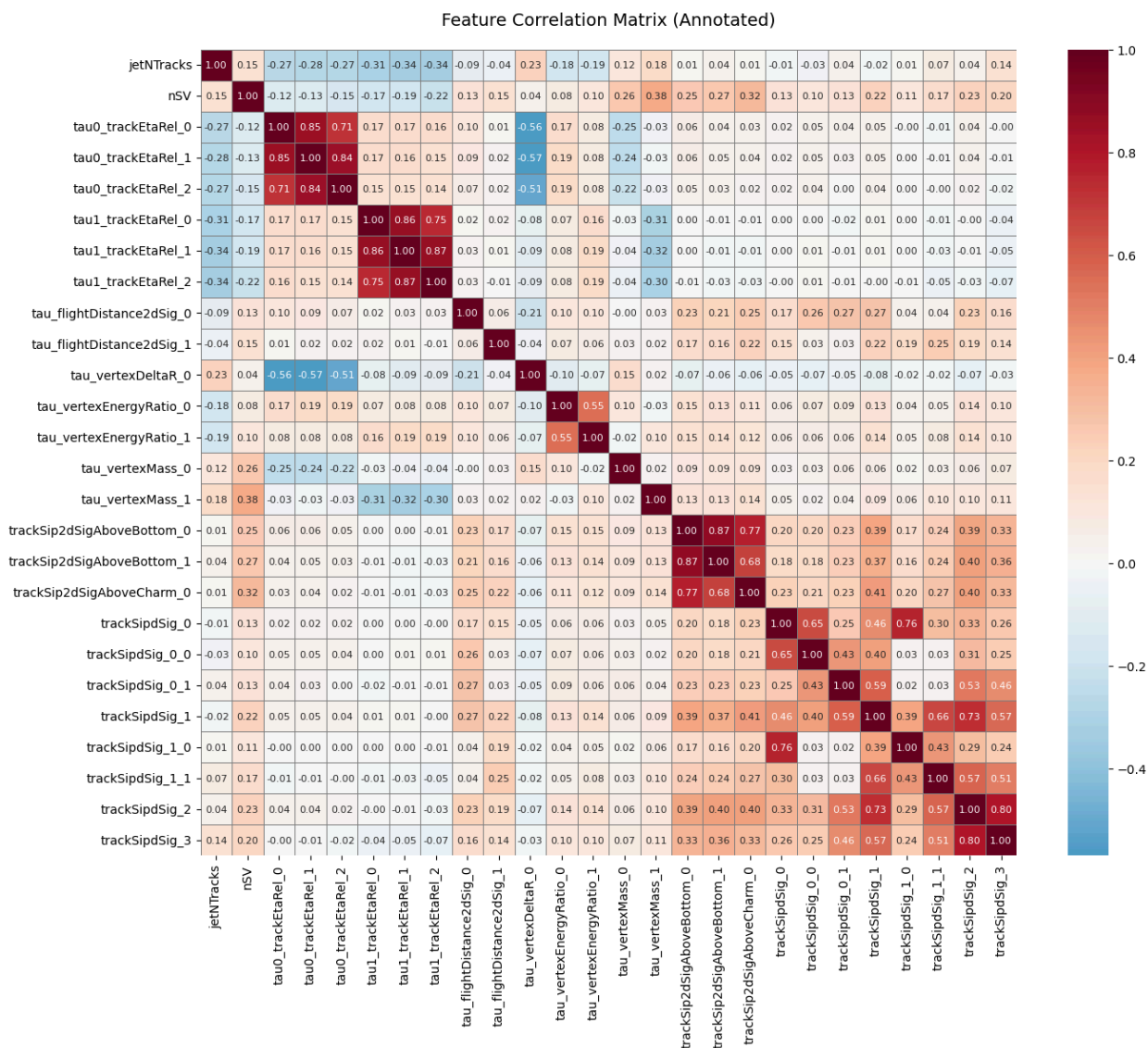


Figure 3: Feature Correlation Matrix

As part of the correlation analysis, a Pearson correlation matrix was computed to identify pairs of variables that exhibit strong linear relationships. The results highlighted several variable pairs with high positive correlations, indicating potential redundancy in the information they carry. Notable examples include:

- tau1_trackEtaRel_1 and tau1_trackEtaRel_2
- trackSip2dSigAboveBottom_0 and trackSip2dSigAboveBottom_1

These features measure similar properties — for instance, track pseudorapidities relative to subjet axes or signed impact parameter significances. Such variables are often closely linked by their physical definitions and the way the CMS detector reconstructs particle tracks.

In machine learning, retaining highly correlated features can lead to multicollinearity, which can negatively affect some algorithms (e.g., Logistic Regression and LDA) by inflating variance in parameter estimates or introducing instability. It may also unnecessarily increase model complexity without improving performance.

To address this:

- One variable from each strongly correlated pair can be removed or deprioritized during model training.
- The retained feature can be chosen based on physical relevance (e.g., stronger association with b-quark signatures), data quality (e.g., fewer missing or extreme values), or predictive value observed during cross-validation.

To illustrate these relationships, 2D histograms were created, showing the joint distribution of values for each pair. These plots help reveal patterns, clusters, or outliers that simple summary statistics may miss.

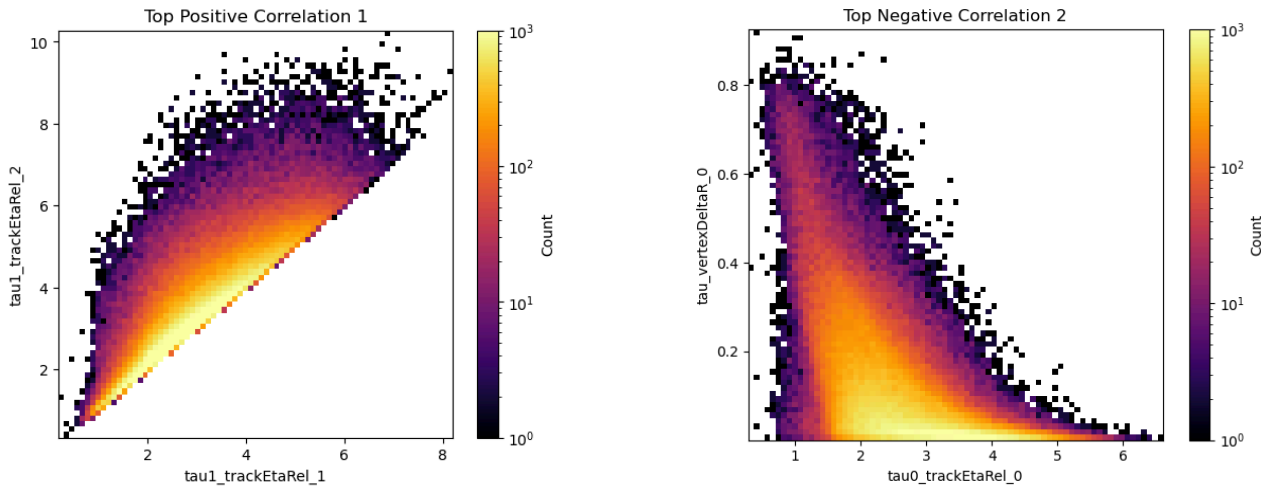


Figure 4: Positive and Negative feature correlation

Principal Component Analysis (PCA)

To gain insight into the internal structure and dominant sources of variation within the signal and background event groups, Principal Component Analysis (PCA) was performed separately on each. Prior to the application of PCA, all numerical features (excluding the classification labels) were standardized to have a mean of 0 and a standard deviation of 1. This standardization ensures that variables with larger scales do not disproportionately influence the principal components.

Following the application of PCA, screen plots were generated for both signal and background datasets.

These plots visualize the proportion of variance explained by each successive principal component.

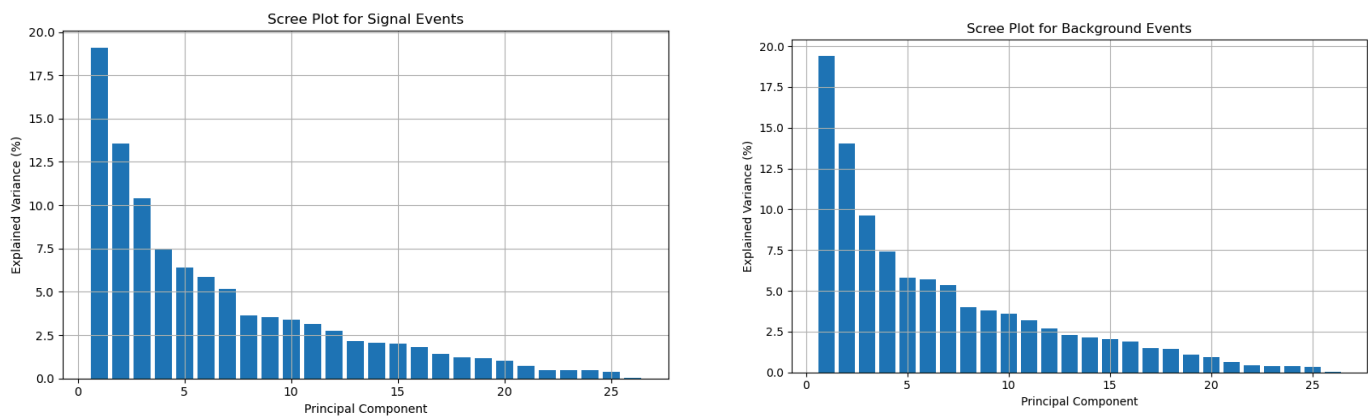


Figure 5: Screen Plots

Screen Events:

The screen plot for signal events indicated that the initial principal components accounted for a substantial portion of the total variance within the data. The discernible drop-off in the explained variance as the component number increased suggested that a reduction in dimensionality could be achieved with minimal loss of salient information.

Background Events:

A similar pattern was observed in the screen plot for background events. However, the distribution of explained variance across the principal components appeared to be more evenly distributed compared to the signal events.

These findings provide valuable information for subsequent steps in the analysis, particularly in determining the feasibility of projecting the high-dimensional data into a lower-dimensional space for visualization purposes or as a pre-processing step prior to the application of machine learning models.

Explained Variance:

The initial principal components demonstrated their capacity to capture a significant fraction of the total variance in both the signal and background datasets. Specifically, the first 10 principal components accounted for approximately 86% of the total variance in the signal event data. A comparable level of variance explanation, around 86.4%, was observed for the first 10 principal components of the background event data.

This observation strongly suggests that the data resides within a lower-dimensional subspace, and consequently, the dimensionality of the dataset can be effectively reduced without incurring a substantial loss of critical information.

Analysis of the component loadings revealed the relative contribution of the original features to each principal component:

- PC1 (Signal and Background): The first principal component in both the signal and background datasets exhibited strong positive correlations with track impact parameter significance variables, including `trackSipdSig_1`, `trackSipdSig_2`, and `trackSip2dSigAboveBottom_0`. These variables are crucial indicators of the displacement of charged particle tracks relative to the primary interaction vertex, a key characteristic for identifying jets originating from b-quarks, which are prevalent in Higgs boson decays.
- PC2 and PC3 (Signal): The second and third principal components for signal events were primarily shaped by pseudorapidity-related variables (belonging to the `tau*_trackEtaRel_*` family) and angular separation features, notably `tau_vertexDeltaR_0`. These features describe the spatial distribution of energy within the jet and the angular separation between decay vertices, providing information that can help differentiate signal jets from background.
- PC2 and PC3 (Background): While the second principal component for background events displayed similar patterns to that of the signal events, the third principal component showed negative contributions from spatial features. This suggests potential differences in the typical spatial topology of background jets compared to signal jets.

The results of the Principal Component Analysis underscore that a limited number of key features, particularly those pertaining to track displacement and jet geometry, encapsulate the majority of the

variance present in both signal and background event data. This finding provides a robust justification for employing dimensionality reduction techniques and for considering the transformed principal components or a selection of the most influential original variables for subsequent classification and modeling tasks.

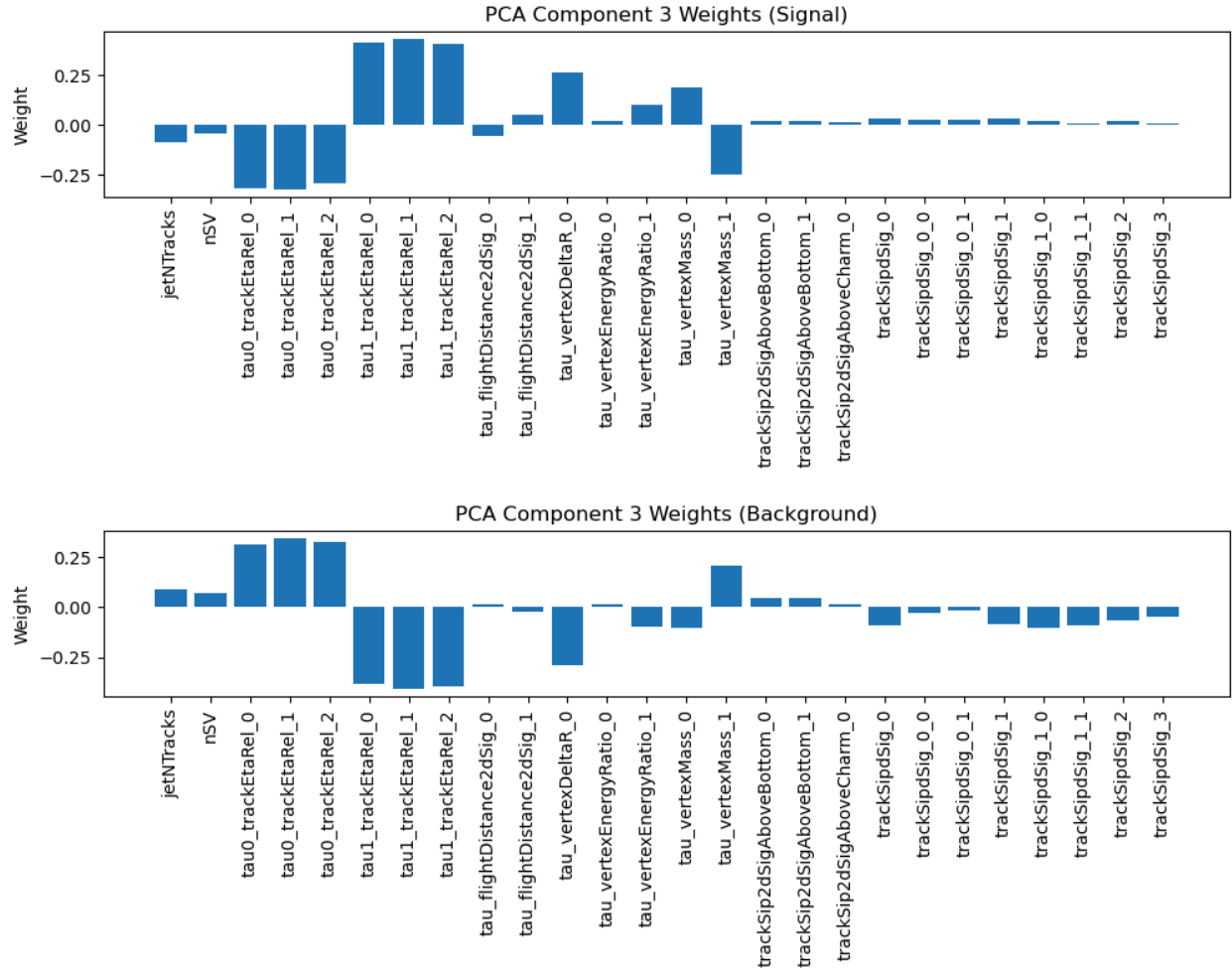


Figure 6: Example Weight plot

2(b) Class Imbalance and Resampling Strategy

An examination of the class distribution within the dataset revealed a notable imbalance:

- Approximately 64.9% of the events were classified as background (isSignal=0).
- Approximately 35.1% of the events were classified as signal (isSignal=1).

This disparity in class proportions can have detrimental effects on the performance of classification models, potentially leading to:

- A bias in the model's predictions towards the majority class (background).
- An artificial inflation of overall accuracy metrics, while simultaneously diminishing the model's capability to accurately identify true signal events.

To address this issue, a random undersampling technique was applied to the background class.

Specifically, the number of background events in the training set was reduced to match the number of signal events. This procedure resulted in a balanced training dataset, ensuring that the classification model treats both classes with equal importance and enhancing its ability to detect the less frequent signal events.

This balancing step is particularly critical in applications within high-energy physics, where rare signal events, such as those originating from Higgs boson decays, can be easily obscured by the more prevalent background noise.

Classification Overview

The classification of signal versus background events was performed using three established machine learning algorithms:

- Naive Bayes
- Linear Discriminant Analysis (LDA)
- Logistic Regression

Two distinct cases were considered in this analysis:

- Case 1: Utilizing all 26 original, raw features.
- Case 2: Utilizing the principal components derived from PCA that collectively explained at least 85% of the total variance in the data.

To mitigate potential model bias arising from the class imbalance, the background class was undersampled to achieve parity with the size of the signal class in the training data. Furthermore, as required by LDA and Logistic Regression, all features were standardized prior to the application of PCA and the training of the classification models.

The performance of the trained models was evaluated using the metrics of precision, recall, and F1-score on a dedicated test dataset.

Model	Case	Accuracy	Signal F1	Background F1
Naive Bayes	All	0.689	0.612	0.740
LDA	All	0.764	0.756	0.771
Logistic Regression	All	0.780	0.771	0.788
Naive Bayes	PCA	0.658	0.559	0.720
LDA	PCA	0.762	0.755	0.768
Logistic Regression	PCA	0.776	0.769	0.784

Key Observations

1. Superior Performance of Logistic Regression

Across both feature sets (original and PCA-reduced), Logistic Regression consistently yielded the highest overall accuracy and the most balanced F1-scores for both signal and background classes. Specifically, it achieved an accuracy of 78.0% when trained on the raw features and 77.6% when trained on the PCA-derived features. This minimal decrease in performance (less than 0.5%) suggests that Logistic Regression is robust to dimensionality reduction in this context.

2. Limitations of Naive Bayes in Signal Detection

While Naive Bayes demonstrated a high recall for background events, its performance in identifying signal events was notably poor, achieving a recall of only 49% in Case 1 and 43% in Case 2. This

imbalance resulted in lower overall F1-scores, rendering Naive Bayes a less suitable choice for applications where the accurate detection of signal events, such as Higgs boson decays, is paramount.

3. Effective Dimensionality Reduction via PCA with Minor Impact on Performance

All three classification algorithms exhibited a slight reduction in performance (typically between 0.5% and 1%) when trained on the PCA-reduced feature set compared to the full set of original features. This observation confirms that PCA effectively preserved a significant portion of the discriminative information present in the data. Consequently, employing PCA for dimensionality reduction appears to be a viable strategy when considerations of computational efficiency or noise reduction are important.

4. Consistent and Reliable Performance of LDA

Linear Discriminant Analysis (LDA) demonstrated a performance level closely comparable to that of Logistic Regression. This makes LDA a strong alternative, particularly in scenarios where the interpretability of the linear decision boundaries offered by the model is a desirable characteristic. In summary, among the evaluated models, Logistic Regression trained on the complete set of original features presented the optimal balance between overall accuracy and precision-recall for both classes, establishing it as the top-performing model for this binary classification task. However, the competitive performance of the PCA-based models indicates that the dataset possesses a strong underlying low-dimensional structure, making dimensionality reduction a feasible option when efficiency is a primary concern.

2(f) Visualising Classification Probabilities

To gain a deeper understanding of the confidence with which the trained models differentiate between signal and background events, the predicted probabilities of signal membership were visualized for both true signal and true background events.

The analysis of these probability distributions revealed two key findings:

- Case 1 (All Features): The probability distributions for signal and background events exhibited a clear separation. Signal events were predominantly assigned probabilities close to 1, while background events were assigned probabilities close to 0. This distinct separation visually reinforces the strong classification performance achieved by the models when trained on the full feature set.

- Case 2 (PCA-Reduced Features): While the probability distributions for signal and background events still showed a degree of separation, there was a noticeable increase in overlap between the two distributions compared to Case 1. This increased overlap suggests that some discriminative information was indeed lost during the dimensionality reduction process performed by PCA, which aligns with the slight decrease in performance metrics observed earlier.

This visualization of predicted probabilities provides a valuable complement to the quantitative accuracy scores, offering insights into the confidence levels associated with the model's classifications and the degree of separation between the predicted classes.

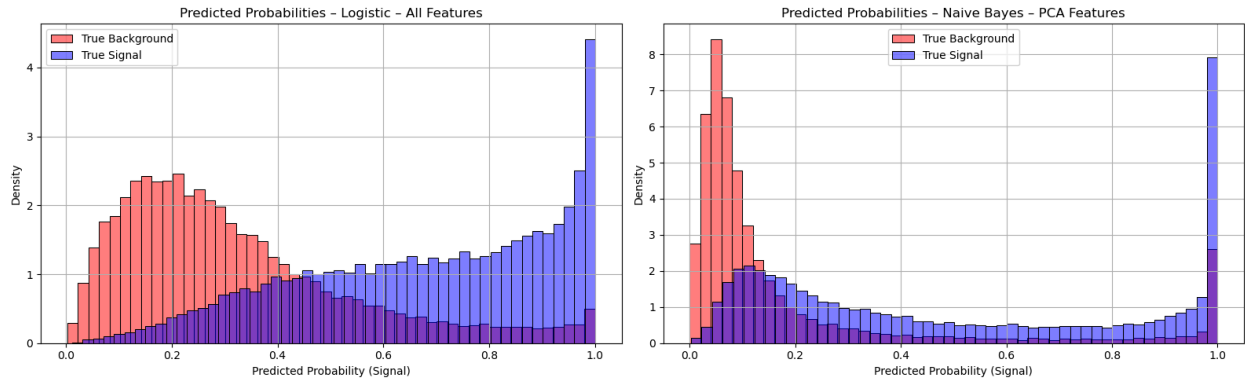


Figure 7: Predicted Probabilities

2(g) Feature Subset Selection and Performance

We used a combination of the following to identify potential subsets:

1. Correlation Analysis (from Section 1e)
 - Highly correlated variable pairs (e.g., tau1_trackEtaRel_1 and tau1_trackEtaRel_2, trackSipdSig_2 and trackSipdSig_3) suggest redundancy — we can drop one from each pair.
2. PCA Loadings (from Section 2a–2b)

- Top principal components for both signal and background were heavily influenced by:
 - trackSipdSig_1, trackSipdSig_2, trackSipdSig_0
 - trackSip2dSigAboveBottom_0, trackSip2dSigAboveCharm_0
 - nSV, tau0_trackEtaRel_1, tau1_trackEtaRel_1

Performance Summary Across All Models

Model	Feature Set	Accuracy	Signal F1	Background F1
Naive Bayes	Full Features	0.689	0.612	0.740
Naive Bayes	PCA Features	0.658	0.559	0.720
Naive Bayes	Subset (9 vars)	0.681	0.607	0.731
LDA	Full Features	0.764	0.756	0.771
LDA	PCA Features	0.762	0.755	0.768
LDA	Subset (9 vars)	0.735	0.722	0.747
Logistic Regression	Full Features	0.780	0.771	0.788

Logistic Regression	PCA Features	0.776	0.769	0.784
Logistic Regression	Subset (9 vars)	0.744	0.733	0.755

Observations:

1. Robust Performance of the Subset Model

The classification models trained on the reduced subset of 9 carefully selected features demonstrated surprisingly robust performance across all algorithms.

Notably, Logistic Regression, when trained on this subset, achieved an accuracy of 74.4%. This represents a decrease of only 3.6 percentage points compared to the model trained on the complete feature set, with a corresponding F1-score for the signal class of 0.733.

2. Consistent Performance of Naive Bayes Across Feature Sets

Naive Bayes consistently exhibited lower performance metrics across all evaluated feature sets (full, PCA-reduced, and the selected subset).

Interestingly, the performance of Naive Bayes when trained on the reduced subset was nearly identical to its performance with the full feature set. This suggests that the predictive capability of Naive Bayes was not significantly impacted by reducing the number of input features.

3. Optimal Performance Trade-off: Full Features or PCA

For achieving the highest overall performance, utilizing the complete set of features or the PCA-reduced feature set remains the optimal strategy. Specifically, Logistic Regression trained on the full feature set continued to be the top-performing model.

However, models trained on the PCA-reduced feature set exhibited a minimal decrease in accuracy (approximately 0.4%) while utilizing a significantly smaller number of components. This indicates that PCA offers a favorable trade-off between dimensionality reduction, computational efficiency, and model performance.

While restricting the input variables to a carefully chosen subset of 9 features did not surpass the performance achieved using the full feature set, it yielded close results, particularly in the case of Logistic Regression. The subset model offers several potential advantages, including:

- Enhanced model interpretability due to the smaller number of input features.
- Reduced computational cost associated with model training and prediction.
- Accuracy and F1-scores that are nearly comparable to those obtained using the full feature set.

These findings suggest that feature selection can be a viable and effective strategy when model simplicity or computational speed is a priority, without incurring a substantial sacrifice in predictive accuracy.

2(h) Threshold Optimisation with Punzi Significance

To improve the performance of our classifier, we optimised the decision threshold using the Punzi significance metric. This metric is designed to balance the trade-off between signal efficiency and background contamination, which is critical in experiments where false positives are highly undesirable. Using the logistic regression model trained on all features, we computed the Punzi significance across a range of probability thresholds. The analysis revealed that the optimal threshold was 0.616. This means that only events with a predicted probability greater than 61.6% should be classified as signal. This is a more conservative threshold than the commonly used default of 0.5, and it helps to reduce the number of background events incorrectly classified as signal.

At this optimal threshold, the maximum Punzi significance achieved was 0.0117, which reflects a strong balance between retaining true signal events and minimising false positives from background. This approach prioritises precision over recall, which aligns with the priorities of particle physics experiments, where a small number of high-quality signal candidates is often more valuable than a large number of potentially contaminated ones.

After identifying the optimal threshold, we applied it to the logistic regression model and compared the results to those obtained using the default threshold. The new threshold resulted in higher precision for the signal class, confirming that the model became more selective in identifying signal-like events.

Although this came at the cost of a slight reduction in recall (fewer signal events were captured), the overall outcome was a more reliable and physics-aligned classifier.

This threshold adjustment illustrates the importance of tuning classification models not just for statistical performance, but for the specific needs and constraints of the domain in which they are applied.

Results at $t = 0.616$

Metric	Background (0)	Signal (1)
Precision	0.699	0.846
Recall (Sensitivity)	0.888	0.617
F1-score	0.782	0.713
Overall Accuracy	0.752	

Applying the optimised threshold of 0.616 improved the precision for signal events to 84.6%, meaning the classifier became more confident and produced fewer false positives. However, recall dropped to 61.7%, as stricter thresholds naturally exclude more true signals. The F1-score for signal decreased slightly to 0.713, reflecting this trade-off.

In high-energy physics, this trade-off is acceptable. Precision is more critical than recall, as false positives from background events can compromise the analysis. A stricter threshold helps ensure that selected events are genuinely signal-like, improving the purity of results.

Results and Conclusions from the Analysis of the CMS Data

This project analysed simulated CMS high-energy collision data with the objective of identifying Higgs boson decays into bottom quarks ($H \rightarrow b\bar{b}$) using machine learning techniques. The dataset contained approximately 225,000 records and 26 numerical features, with two classes: signal (Higgs decay) and background (QCD jets).

During the data exploration phase, no critical missing values or placeholder values such as -999 or -1 were found, apart from a few non-impactful entries. Variables such as the number of secondary vertices (nSV), track impact parameter significance metrics (trackSipdSig_*), and jet shape variables

(tau_trackEtaRel_*) were observed to be informative for distinguishing signal from background.

Correlation analysis revealed several highly correlated variable pairs, suggesting potential redundancy.

Principal Component Analysis (PCA) confirmed that the first five to six principal components accounted for approximately 60–65% of the total variance. The top contributing features in both signal and background events included nSV, trackSip*, and tau* variables, reinforcing their importance in detecting bottom quark jets.

Three classification algorithms—Naive Bayes, Linear Discriminant Analysis (LDA), and Logistic Regression—were trained and evaluated under three different feature configurations: using all features, using PCA-transformed components, and using a manually selected subset of nine features. Logistic Regression performed best with all features, achieving 78.0% accuracy and an F1-score of 0.771 for signal. The PCA-reduced setup delivered slightly lower accuracy (77.6%) but offered reduced dimensionality. The feature subset model achieved 74.4% accuracy, suggesting that simpler models can still offer competitive performance with enhanced interpretability and efficiency.

To further optimise model performance, the decision threshold for the logistic regression classifier was adjusted using the Punzi significance metric, which is particularly relevant in high-energy physics where controlling false positives is critical. The optimal threshold of 0.616 improved the model's precision for identifying signal events to 84.6%, though recall decreased to 61.7%. This trade-off resulted in a more conservative and reliable classifier, better suited for detecting rare physics events like Higgs decays in a background-dominated environment.

Literature Review

The methodology and conclusions of this analysis were guided by both domain knowledge and insights from the following key sources:

- CMS Collaboration (2020) – Identification of heavy-flavour jets with the CMS detector in pp collisions at 13 TeV (JINST 13, 2018, P05011) explains the physical basis for using features such as secondary vertices and impact parameters in b-jet tagging.
- Punzi, G. (2003) – Sensitivity of searches for new signals and its optimization (arXiv:physics/0308063) introduces the Punzi significance metric used to optimise threshold selection in rare-event classification.
- Bourilkov, D. (2014) – Machine Learning for High Energy Physics (arXiv:1409.8435) justifies the choice of interpretable classifiers such as Logistic Regression and LDA in high-energy physics contexts.
- James, G. et al. (2013) – An Introduction to Statistical Learning (Springer) provides theoretical foundations for PCA, classification algorithms, and model evaluation metrics applied in this study.
- Aad, G. et al. (ATLAS Collaboration, 2020) – Machine Learning in the ATLAS Experiment (JINST 15, 2020, P05006) illustrates practical use of ML and feature selection in real collider experiments, reinforcing the importance of precision and dimensionality reduction in such settings.
- Paperspace (2021) – Top 4 Techniques for Handling the Missing Values in Machine Learning discusses industry-standard practices for identifying and addressing missing or placeholder values in machine learning workflows, which informed the initial data cleaning step of this analysis.

The analysis demonstrates that physics-informed features, such as those based on b-tagging, are highly effective in distinguishing signal from background in particle collision data. Both PCA and manual feature selection can reduce dimensionality with minimal performance loss. Finally, threshold tuning via

the Punzi significance proves essential when high precision is required, as is often the case in high-energy physics. The final logistic regression model, enhanced by an optimised threshold, strikes a strong balance between precision, recall, and scientific reliability.